Grade-12 (Formulae)

Inverse Trigonometry

(1)

Function	Domain (x)	Range (y)
$\sin^{-1} x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\operatorname{cosec}^{-1} x$	$R - (-1, 1)$	$[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$
$\sec^{-1} x$	$R - (-1, 1)$	$[0, \pi] - \{\frac{\pi}{2}\}$
$\tan^{-1} x$	R	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$\cot^{-1} x$	R	$(0, \pi)$.

(2)

$$\sin^{-1} \frac{1}{x} = \operatorname{cosec}^{-1} x$$

$$\cos^{-1} \frac{1}{x} = \sec^{-1} x$$

$$\tan^{-1} \frac{1}{x} = \begin{cases} \cot^{-1} x & x > 0 \\ -\pi + \cot^{-1} x & x < 0 \end{cases}$$

(3)

$$\sin^{-1}(-x) = -\sin^{-1} x$$

$$\cos^{-1}(-x) = \pi - \cos^{-1} x$$

$$\tan^{-1}(-x) = -\tan^{-1} x$$

$$\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$$

$$\sec^{-1}(-x) = \pi - \sec^{-1} x$$

$$\cot^{-1}(-x) = \pi - \cot^{-1} x$$

(4)

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$$

$$\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}$$

$$\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$$

(5)

$$\tan^{-1} x + \tan^{-1} y = \begin{cases} \tan^{-1}\left(\frac{x+y}{1-xy}\right) & \text{if } xy < 1 \\ \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right) & \text{if } x > 0,\ y > 0 \text{ and } xy > 1 \\ -\pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right) & \text{if } x < 0,\ y < 0,\ xy > 1 \end{cases}$$

$$\tan^{-1} x - \tan^{-1} y = \begin{cases} \tan^{-1}\left(\frac{x-y}{1+xy}\right) & xy > -1 \\ \pi + \tan^{-1}\left(\frac{x-y}{1+xy}\right) & \text{if } x > 0,\ y < 0 \text{ and } xy < -1 \\ -\pi + \tan^{-1}\left(\frac{x-y}{1+xy}\right) & \text{if } x < 0,\ y > 0 \text{ and } xy < -1 \end{cases}$$

(6) $$2\tan^{-1} x = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$$

i.e. $$2\tan^{-1} x = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

Matrices

Transpose properties

1. $(A')' = A$

   $(AB)' = B'A'$

   $(A \pm B)' = A' \pm B'$

   $(KA)' = KA'$

   $(ABC)' = C'B'A'$

2. A is symmetric matrix

   then $A' = A$

   If A is skew symmetric matrix

   then $A' = -A$.

3. Any matrix A can be expressed as sum of symmetric and skew symmetric matrices.

   $$A = P + Q$$

   where $P = \frac{1}{2}(A + A')$

   and $Q = \frac{1}{2}(A - A')$

Properties of adjoint matrix.

1. $A(\text{adj} A) = (\text{adj} A)A = |A| I$
2. $|\text{adj} A| = |A|^{n-1}$ where n is order of A.
3. $\text{adj}(KA) = K^{n-1}(\text{adj} A)$ where n is order of A and K is real number

Properties of inverse matrix

$$A^{-1} = \frac{1}{|A|}(\text{adj} A)$$

1. $AB = BA = I$ then $B = A^{-1}$ and $A = B^{-1}$
2. $(A^{-1})^{-1} = A$
3. $(AB)^{-1} = B^{-1}A^{-1}$.

Elementary transformation

1. Interchange two rows or coloumn $\begin{pmatrix} R_i \leftrightarrow R_j \\ C_i \leftrightarrow C_j \end{pmatrix}$
2. $R_i \rightarrow K R_i$ or $C_i \rightarrow K C_i$
3. $R_i \rightarrow R_i + KR_j$ or $C_i \rightarrow C_i + KC_j$

Finding inverse using elementary transformation.

1. use Row operation only then $A = IA$
2. use Coloumn operation only then $A = AI$

Solution of system of linear equations

$$a_1 x + b_1 y + c_1 z = d_1$$
$$a_2 x + b_2 y + c_2 z = d_2$$
$$a_3 x + b_3 y + c_3 z = d_3$$

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

$$AX = B$$

then $X = A^{-1} B$

$$X = \frac{1}{|A|} (\text{adj} A)(B)$$

Case I

If $|A| \neq 0$ then it has consistent and unique solⁿ.

Case II if $|A| = 0$ and $(\text{adj} A) B = 0$ then it has many infinite solⁿ and consistent

Case III if $|A| = 0$ and $(\text{adj} A) B \neq 0$ then it is inconsistent and no solution

Determinants

Properties of Determinants

(1) Value of determinant remains unchanged if coloumn and rows are interchanged.

(2) If any two rows or column are interchanged then value of determinant becomes negative of previous one.

(3) If any two rows or column are proportional then determinant = 0.

(4) If any two rows or column are same then det. = 0

(5)
$$\begin{vmatrix} a_1+a_2 & b_1 & c_1 \\ p_1+p_2 & b_2 & c_2 \\ q_1+q_2 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ p_1 & b_2 & c_2 \\ q_1 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} a_2 & b_1 & c_1 \\ p_2 & b_2 & c_2 \\ q_2 & b_3 & c_3 \end{vmatrix}$$

(6)
$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

if $R_1 \to KR_1$ then $\frac{1}{K}\begin{vmatrix} Ka_1 & Kb_1 & Kc_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

(7) $R_2 \to R_2 + KR_1$

or

$C_2 \to C_2 - KC_1$

8. $|AB| = |A||B|$

9. $|A|^n = |A^n|$

10. Minor of element $a_{11} = \begin{vmatrix} b_{22} & c_{23} \\ b_{32} & c_{33} \end{vmatrix} = M_{11}$

11. Cofactor of element $a_{ij} = C_{ij}$ or $A_{ij} = (-1)^{i+j} M_{ij}$

12.
$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\Delta = a_{11} C_{11} + a_{12} C_{12} + a_{13} C_{13}$$

but $a_{11} C_{31} + a_{12} C_{32} + a_{13} C_{33} = 0$

Continuity and differentiability

1. A $f(x)$ is continuous at $x = a$ if

$$L \cdot H \cdot L = R \cdot H \cdot L = f(a)$$

or

$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = f(a)$$

or

$$\lim_{h \to 0} f(a-h) = \lim_{h \to 0} f(a+h) = f(a)$$

2. A f(x) is said to be differentiable if

$$L \cdot H \cdot D = R \cdot H \cdot D = \text{finite value}$$

$$f'(a^-) = f'(a^+)$$

$$L \cdot H \cdot D = f'(a^-) = \lim_{h \to 0} \frac{f(a-h) - f(a)}{-h}$$

$$R \cdot H \cdot D = f'(a^+) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

3. Logarithm properties

$$y = \log_a x \Rightarrow x = a^y$$

we use base 'e' for differentiation

$$y = \log_e x = \ln x$$

a) $\log\left(\frac{a}{b}\right) = \log a - \log b$

b) $\log(ab) = \log a + \log b$

c) $\log a^m = m \log a$

d) $\log 1 = 0$

e) $\log_a a = 1$

f) Base changing property $\log_a b = \frac{\log_c b}{\log_c a}$

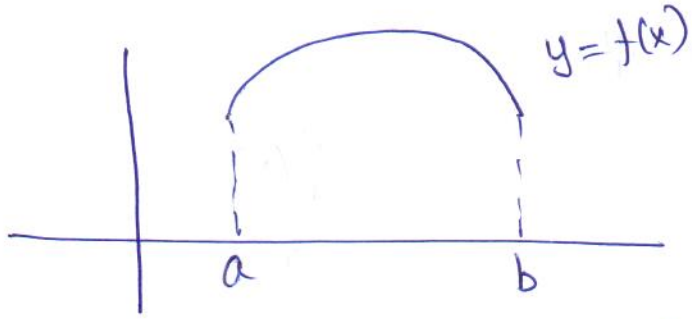
g) $a^{\log_a b} = b$ eg:- $2^{\log_2 3} = 3$

(4) Differentiation formula.

	$f(x)$	$f'(x)$
a)	$\sin x$	$\cos x$
b)	$\cos x$	$-\sin x$
c)	$\tan x$	$\sec^2 x$
d)	$\sec x$	$\sec x \tan x$
e)	$\cot x$	$-\operatorname{cosec}^2 x$
f)	$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
g)	$\log_e x$	$\frac{1}{x}$
h)	e^x	e^x
i)	$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$
j)	$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}$
k)	$\tan^{-1} x$	$\frac{1}{1+x^2}$
l)	$\cot^{-1} x$	$\frac{-1}{1+x^2}$
m)	$\sec^{-1} x$	$\frac{1}{x\sqrt{x^2-1}}$
n)	$\operatorname{cosec}^{-1} x$	$\frac{-1}{x\sqrt{x^2-1}}$

(5)

(5) Rolle's theorem



If $f(x)$ is cont on $[a, b]$
& diff. on (a, b)
& $f(a) = f(b)$

then $f'(c) = 0$ where $c \in (a, b)$

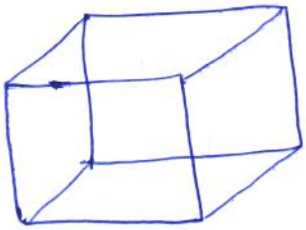
(6) Mean value theorem

If $f(x)$ is cont. on $[a, b]$
and diff. on (a, b)

then $f'(c) = \dfrac{f(b) - f(a)}{b - a}$ where $c \in (a, b)$.

Application of derivatives

(1) Rate of change of function

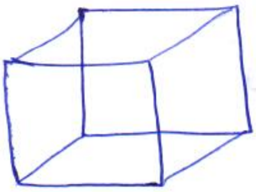


Cuboid

$C.S.A = L.S.A = 2(l+b)h$

$T.S.A = 2(lb + bh + hl)$

$\text{volume} = l \times b \times h$



Cube

$l = b = h = a$

$C.S.A = 4a^2$

$T.S.A = 6a^2$

$\text{volume} = a^3$

Cylinder

$C.S.A = 2\pi r h$

$T.S.A = 2\pi r(h+r)$

$\text{volume} = \pi r^2 h$

Cone

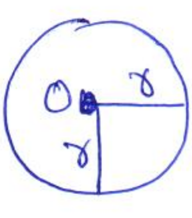
$C.S.A = \pi r l$

$T.S.A = \pi r(l+r)$

$\text{volume} = \frac{1}{3}\pi r^2 h$

where $l^2 = b^2 + h^2$.

sphere



$T.S.A = 4\pi r^2$

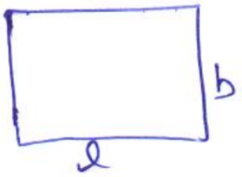
$\text{volume} = \frac{4}{3}\pi r^3$

hemisphere

$C.S.A = 2\pi r^2$

$T.S.A = 3\pi r^2$

$\text{volume} = \frac{2}{3}\pi r^3$



sectangle

Perimeter $= 2(l+b)$

Area $= l \times b$

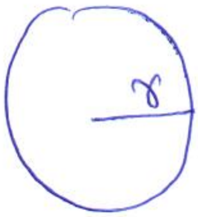
Square

Perimeter $= 4\,(\text{side}) = 4a$

Area $= (\text{side})^2 = a^2$

→ Marginal cost $= \dfrac{d}{dx}(\text{Total cost})$

→ marginal revenue $= \dfrac{d}{dx}(\text{total revenue})$



Circle

Circumference $= 2\pi r$

Area $= \pi r^2$

2 Increasing and decreasing functions

a) If $f'(x) > 0$ then $f(x)$ is strictly $\uparrow$ing.

b) If $f'(x) \geq 0$ then $f(x)$ is $\uparrow$ing.

c) If $f'(x) < 0$ then $f(x)$ is strictly $\downarrow$ing.

d) If $f'(x) \leq 0$ then $f(x)$ is $\downarrow$ing.

How to find increasing & decreasing interval of function

Step I find $f'(x)$ and write into linear factors

Step II Put $f'(x) = 0$, Both N^r & $D^r = 0$ to find cricitical pts. & represent critical pts. on number lin

Step III Then find scheme and mention the interval about monotonoity of function.

③ Tangents and normals

a) slope of tangent, $m_T = \left.\frac{dy}{dx}\right|_{(x_0, y_0)} = \tan\theta$ (where θ is angle of inclination)

b) slope of normal, $m_N = \frac{-1}{m_T} = -\left.\frac{dx}{dy}\right|_{(x_0, y_0)}$

where (x_0, y_0) is point of contact.

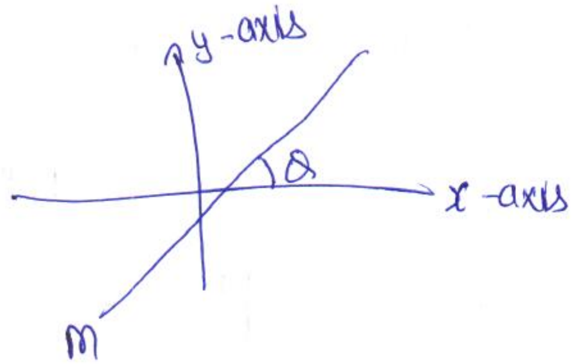
c) Equation of tangent

$$y - y_0 = m_T (x - x_0)$$

d) Equation of normal

$$y - y_0 = m_n (x - x_0)$$

e) slope of any line $= m = \tan\theta$



f) If two lines are perpendicular then $m_1 m_2 = -1$

g) If two lines are parallel then $m_1 = m_2$.

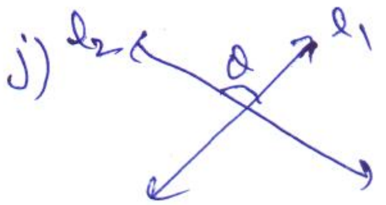
h) If line is parallel to x-axis then $m = 0$.

g) If line is perpendicular to x-axis then $m =$ not defined.

i) If $m_T = 0$ then $m_n =$ not defined

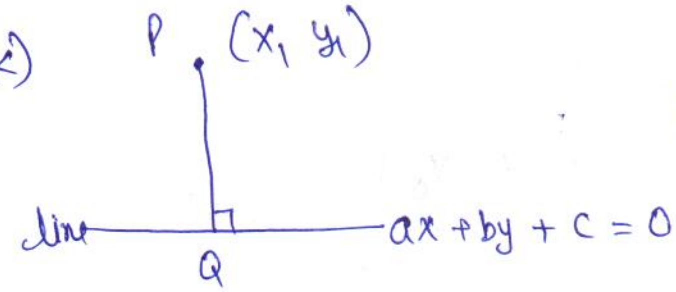
and

If $m_T =$ not defined then $m_n = 0$

j) 

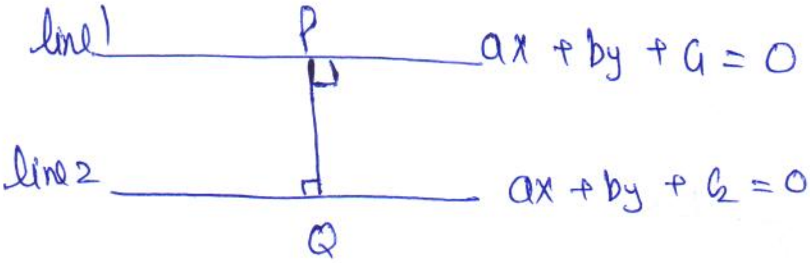
angle between two lines,

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

k) 

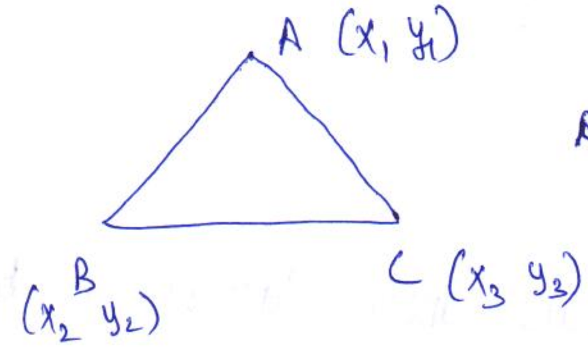
$$PQ = \left|\frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}}\right|$$

l)



$$PQ = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}.$$

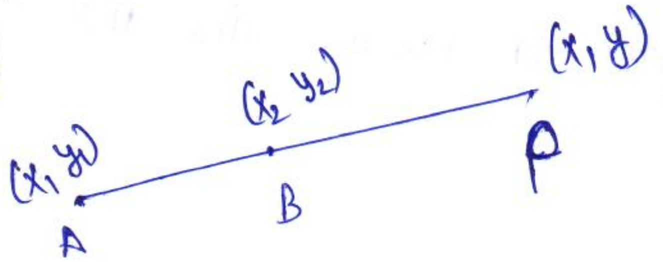
k)



$$\text{Area of } \Delta ABC = \left| \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \right|$$

If pts. are collinear then area of triangle = 0

m)



If two pts. are on same line

then equation line AB can be find using determinal

concept

$$\text{Area of } \Delta ABP = \frac{1}{2} \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

Simplify above to get eq^n

in terms of 'x' and 'y'.

4 Approximations

$$\Delta y = \frac{dy}{dx} \cdot \Delta x$$

$$f(x + \Delta x) = f(x) + \Delta y$$

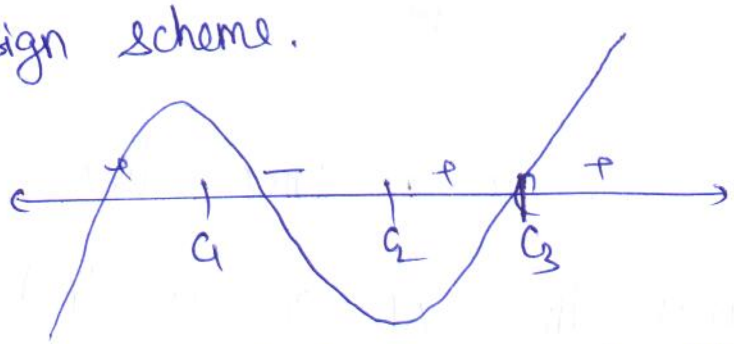
5 maxima and minima

a) First derivative Test

Step I find $f'(x)$ and make linear factors in both numerator and denominator.

Step II Put each linear factor $= 0$ to find critical pts.

Step III Put critical pts. on number line and find sign scheme.



$c_1 \rightarrow$ pt. of local max. and $f(c_1) \rightarrow$ local max. value

$c_2 \rightarrow$ pt. of local minima and $f(c_2) \rightarrow$ local min. value.

$c_3 \rightarrow$ pt. of inflection (no local max or local min)

b) Second derivative Test

Step I find $f'(x)$ and make linear factors.

Step II Put each linear factor $= 0$ in both N^r and D^r. Let c_1, c_2, c_3 be critical pts.

Step III find $f''(x)$

$f''(c_1) > 0$ then $c_1 \rightarrow$ pt of local min.
$f(c_1) \rightarrow$ local min. value

$f''(c_2) < 0$ then $c_2 \rightarrow$ pt. of local max.
$f(c_2) \rightarrow$ local max. value

$f''(c_3) = 0$ then Test fails go to first derivative test for this point.

c) How to find Global (absolute) maximum or minimum.

$$y = f(x) \quad \forall \quad x \in [a, b]$$

Step I find $f'(x)$ and make linear factors

Step II Put each factor $= 0$ to find critical pts.
Let c_1, c_2 be critical pts.

Step III find value of $f(x)$ at critical pts. and at end pts. 'a' and 'b'.
Examine the maximum and minimum value from above values.

Integration

(a) function $f(x)$ — Integral or anti-derivative $\int f(x)dx$

a) $\int x^n dx = \frac{x^{n+1}}{n+1} + c$, $n \neq -1$

b) $\int \frac{1}{x} dx = \log_e x + c$

c) $\int e^x dx = e^x + c$

d) $\int a^x dx = \frac{a^x}{\ln a} + c$

e) $\int \sin x \, dx = -\cos x + c$

f) $\int \cos x \, dx = \sin x + c$

g) $\int \sec^2 x \, dx = \tan x + c$

h) $\int \operatorname{cosec}^2 x \, dx = -\cot x + c$

i) $\int \sec x \tan x \, dx = \sec x + c$

j) $\int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + c$

k) $\int \cot x \, dx = \log|\sin x| + c$

l) $\int \tan x \, dx = \log|\sec x| + c$

i) $\int \sec x \, dx = \log|\sec x + \tan x| + c$

m) $\int \operatorname{cosec} x \, dx = \log|\operatorname{cosec} x - \cot x| + c$

n) $\int \frac{dx}{\sqrt{a^2 - x^2}} = \frac{1}{a} \sin^{-1}\left(\frac{x}{a}\right) + c$

o) $\int \frac{-dx}{\sqrt{a^2 - x^2}} = \frac{1}{a} \cos^{-1}\left(\frac{x}{a}\right) + c$

p) $\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + c = -\frac{1}{a} \cot^{-1}\left(\frac{x}{a}\right) + c$

q) $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \sec^{-1}\left(\frac{x}{a}\right) + c = -\frac{1}{a} \operatorname{cosec}^{-1}\left(\frac{x}{a}\right) + c$

Trigonometric rules useful for integration

a) $\cancel{\cos^2 x}$ $\cos^2 A = \frac{1 + \cos 2A}{2}$

b) $\sin^2 A = \frac{1 - \cos 2A}{2}$

c) $\sin 3A = 3\sin A - 4\sin^3 A$

↳ you can find $\sin^3 A$ from here.

d) $\cos 3A = 4\cos^3 A - 3\cos A$

↳ you can find $\cos^3 A$ from here.

e) $2\sin A \cos B = \sin(A+B) + \sin(A-B)$

$2\cos A \cos B = \cos(A+B) + \cos(A-B)$

$-2\sin A \sin B = \cos(A+B) - \cos(A-B)$

Standard substitution for Integration

function	Substitution
a) (x^2+a^2), $\sqrt{a^2+x^2}$, $\frac{1}{\sqrt{a^2+x^2}}$	$x = a\tan\theta$, $a\cot\theta$,
b) (a^2-x^2), $\sqrt{a^2-x^2}$, $\frac{1}{\sqrt{a^2-x^2}}$	$x = a\sin\theta$ or $a\cos\theta$ according to requirement
c) $\left(x \pm \sqrt{x^2 \pm a^2}\right)^n$	Put expression inside the bracket $= t$
d) (x^2-a^2), $\sqrt{x^2-a^2}$, $\frac{1}{\sqrt{x^2-a^2}}$	$x = a\sec\theta$ or $a\operatorname{cosec}\theta$
e) $\sqrt{\frac{a-x}{a+x}}$ or $\sqrt{\frac{a+x}{a-x}}$	Put $x = a\cos 2\theta$
f) $\sqrt{\frac{x}{a+x}}$ or $\sqrt{\frac{a+x}{a}}$	$x = a\tan^2\theta$ or $a\cot^2\theta$
g) $\sqrt{\frac{x}{a-x}}$ or $\sqrt{\frac{a-x}{x}}$	$x = a\sin^2\theta$ or $a\cos^2\theta$

Special integrals

a) $\displaystyle\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + c$

b) $\displaystyle\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right| + c$

c) $\displaystyle\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right| + c$

d) $\displaystyle\int \frac{dx}{\sqrt{a^2-x^2}} = \frac{1}{a}\sin^{-1}\left(\frac{x}{a}\right) + c$

e) $\displaystyle\int \frac{dx}{\sqrt{x^2-a^2}} = \log\left|x+\sqrt{x^2-a^2}\right| + c$

f) $\displaystyle\int \frac{dx}{\sqrt{x^2+a^2}} = \log\left|x+\sqrt{x^2+a^2}\right| + c$

g) $\displaystyle\int \sqrt{x^2+a^2}\,dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\log\left|x+\sqrt{x^2+a^2}\right| + c$

h) $\displaystyle\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\log\left|x+\sqrt{x^2-a^2}\right| + c$

i) $\displaystyle\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + c$

Important forms to be converted into special Integrals

Form – I

$$\int (px+q)\sqrt{ax^2+bx+c}\;dx$$

write,

$$px+q = A\left(ax^2+bx+c\right)' + B$$

find A and B first

then

$$(px+q)\sqrt{ax^2+bx+c} = A(2ax+b)\sqrt{ax^2+bx+c} + B\sqrt{ax^2+bx+c}$$

$$\int (px+q)\sqrt{ax^2+bx+c}\;dx = A\int (2ax+b)\sqrt{ax^2+bx+c}\;dx + B\int\sqrt{ax^2+bx+c}\;dx$$

$\downarrow$ let $ax^2+bx+c = t$ and then Integrate

$\downarrow$ integrate using completing the square method.

Form - II

$$\int \frac{1}{ax^2+bx+c}\,dx \quad \text{or} \quad \int \frac{dx}{\sqrt{ax^2+bx+c}} \quad \text{or} \quad \int \sqrt{ax^2+bx+c}\,dx$$

integrate, using completing the square method and then appropriate formula.

Form - III

$$\int \frac{px+q}{ax^2+bx+c}\,dx \quad \text{or} \quad \int \frac{px+q}{\sqrt{ax^2+bx+c}}\,dx$$

Put

$$px+q = A\,(ax^2+bx+c)' + B$$

Find A and B then by comparing.

$$\int \frac{px+q}{ax^2+bx+c}\,dx = \int \frac{A\,(2ax+b)}{ax^2+bx+c}\,dx + B\int \frac{dx}{ax^2+bx+c}$$

$\downarrow$ Put $ax^2+bx+c = t$ and then integrate

$\downarrow$ use completing the square method.

Form V

$$\int \frac{x^2+a^2}{x^4+Kx^2+a^2}\,dx \quad \text{or} \quad \int \frac{x^2-a^2}{x^4+Kx^2+a^2}\,dx$$

first divide by x^2 in N^r, D^r

eg:- $$I = \int \frac{x^2+4}{x^4+2x^2+16}\,dx$$

Div by x^2 in N^r, D^r

$$I = \int \frac{\left(1+\frac{4}{x^2}\right)dx}{\left(x^2+\frac{16}{x^2}\right)+2}$$

completing the square

$$= \int \frac{\left(1+\frac{4}{x^2}\right)dx}{\left(x-\frac{4}{x}\right)^2+8+2}$$

$$= \int \frac{\left(1+\frac{4}{x^2}\right)dx}{\left(x-\frac{4}{x}\right)^2+\left(\sqrt{10}\right)^2}$$

Put $x-\frac{4}{x}=t$

$$\left(1+\frac{4}{x^2}\right)dx = dt$$

$$I = \int \frac{dt}{t^2+\left(\sqrt{10}\right)^2}$$

use formula now

See, more question from notes.

Form VI

$$\int \frac{dx}{a + b\sin^2 x}, \quad \int \frac{dx}{a + b\cos^2 x} \;\&\; \int \frac{dx}{a\sin^2 x + b\cos^2 x}$$

$$\int \frac{dx}{a\sin^2 x + b\cos^2 x + c}, \quad \int \frac{dx}{(a\sin x + b\cos x)^2}$$

Step I Divide by $\cos^2 x$ & $\sin^2 x$ in N^r, D^r.

Step II Replace $\sec^2 x$, if any in D^r by $(1 + \tan^2 x)$

Step III Replace $\tan x = t$ then $\sec^2 x \, dx = dt$

Form VII

$$\int \frac{dx}{a + b\sin x}, \quad \int \frac{dx}{a + b\cos x}, \quad \int \frac{dx}{a\sin x + b\cos x}, \quad \int \frac{dx}{a\sin x + b\cos x + c}$$

Step I Put $\sin x = \frac{2\tan x/2}{1 + \tan^2 x/2}$ and $\cos x = \frac{1 - \tan^2 x/2}{1 + \tan^2 x/2}$

Step II Replace $1 + \tan^2 \frac{x}{2}$ by $\sec^2 \frac{x}{2}$ in N^r after taking L.C.M in D^r.

Step III Put $\tan \frac{x}{2} = t$ and then $\frac{1}{2}\sec^2 \frac{x}{2}\, dx = dt$

Form VIII

$$\int \frac{a\sin x + b\cos x}{c\sin x + d\cos x}\, dx$$

write,

$$N^r = A\,(D^r)' + B\,(D^r)$$

Find A and B first by comparing, then

so, $(a\sin x + b\cos x) = A(c\cos x - d\sin x) + B(c\sin x + d\cos x)$

$$\int \frac{a\sin x + b\cos x}{c\sin x + d\cos x}\, dx = A\int \frac{c\cos x - d\sin x}{c\sin x + d\cos x}\, dx + B\int dx$$

Form IX

$$\int \frac{a\sin x + b\cos x + c}{p\cos x + q\sin x + r}\, dx$$

write,

$$N^r = A\,(D^r)' + B\,(D^r) + C$$

Find A, B and C by comparing, then

$$\int \frac{(a\sin x + b\cos x + c)}{p\cos x + q\sin x + r}\, dx = A\int \frac{p(-\sin x) + q\cos x}{p\cos x + q\sin x + r}\, dx + B\int dx + C\int \frac{dx}{p\cos x + q\sin x + r}$$

Put $D^r = t$

Integrate using form VII

Form X

$$\int \frac{x^2+1}{x^4+\lambda x^2+1}\,dx \quad \text{or} \quad \int \frac{x^2-1}{x^4+\lambda x^2+1}\,dx \quad \text{or} \quad \int \frac{dx}{x^4+\lambda x^2+1}$$

Step I Divide N^r, D^r by x^2.

Step II Express D^r in form of $\left(x \pm \frac{1}{x}\right)^2 \pm K^2$

Step III Introduce $\left(1+\frac{1}{x^2}\right)$ or $\left(1-\frac{1}{x^2}\right)$ or both in N^r

Step IV Put $\left(x+\frac{1}{x}\right)=t$ or $\left(x-\frac{1}{x}\right)=t$ as the case may b

Step V Integral reduced to form $\int \frac{1}{x^2+a^2}\,dx$ or $\int \frac{dx}{x^2-a^2}$

Form XI

$$\int \sqrt{\tan x}\,dx\,, \quad \int \sqrt{\cot x}\,dx\,, \quad \int \frac{dx}{\sin^4 x+\cos^4 x}\,, \quad \int \frac{dx}{\sin^6 x+\cos^6 x}$$

$$\int \frac{\pm \sin x \pm \cos x}{a+b\sin x\,.\cos x}\,dx\,.$$

Put $\tan x = t^2 \Rightarrow \sec^2 x\,dx = 2t\,dt$

$$\Rightarrow \int \frac{2t^2}{t^4+1}\,dt = \int \frac{t^2+1}{t^4+1}\,dt + \int \frac{t^2-1}{t^4+1}\,dt$$

now same as form X.

Integration by Parts) (ILATE)

$$\int \underset{I}{f(x)}\,\underset{II}{g(x)}\,dx = f(x)\int g(x)\,dx - \int\left[\frac{d\,f(x)}{dx}\cdot\int g(x)\,dx\right]dx.$$

(i) Prefer order of fun^c $I \rightarrow L \rightarrow A \rightarrow T \rightarrow E$

(ii) If one of the fun^c is not directly integrable, then we take it as one function.

(iii) If only one fun^c is there i.e $\int \log x\,dx$ then 1 (unity) is taken as second function.

(iv) If both fun^c are integrable, then the first function is chosen in such a way that its derivative vanishes easily or the fun^c obtained in integral sign is easily vanishable.

Integral of form

$$\int e^x\left(f(x) + f'(x)\right)dx = e^x f(x) + C$$

eg:- $\displaystyle\int e^x \frac{x}{(x+1)^2}\,dx = \int e^x\left[\frac{(x+1)-1}{(x+1)^2}\right]dx$

$$= \int e^x\left[\underset{f(x)}{\underset{\uparrow}{\frac{1}{x+1}}} - \underset{f'(x)}{\underset{\uparrow}{\frac{1}{(x+1)^2}}}\right]dx$$

Integration using partial fraction (applied on proper rational fu

① $\dfrac{px+q}{(x-a)(x-b)} \Rightarrow \dfrac{A}{x-a} + \dfrac{B}{x-b}$

② $\dfrac{px^2+qx+r}{(x-a)(x-b)(x-c)} = \dfrac{A}{x-a} + \dfrac{B}{x-b} + \dfrac{C}{x-c}$

③ $\dfrac{px+q}{(x-a)^3} = \dfrac{A}{x-a} + \dfrac{B}{(x-a)^2} + \dfrac{C}{(x-a)^3}$

④ $\dfrac{px^2+qx+r}{(x-a)^2(x-b)} = \dfrac{A}{x-a} + \dfrac{B}{(x-a)^2} + \dfrac{C}{(x-b)}$

⑤ $\dfrac{px^2+qx+r}{(x-a)(x^2+bx+c)} = \dfrac{A}{x-a} + \dfrac{Bx+C}{x^2+bx+c}$

↳ when (x^2+bx+c) can-not be factorized.

⑥ $\dfrac{px^3+qx^2+rx+d}{(x^2+ax+b)(x^2+cx+d)} = \dfrac{Ax+B}{x^2+ax+b} + \dfrac{Cx+D}{x^2+cx+d}$

where (x^2+ax+b) and (x^2+cx+d) cannot be factorized

Properties of Definite Integral

(a) $\int_a^b f(x)\,dx = \Big[F(x)\Big]_a^b = F(b) - F(a)$

(b) $\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$

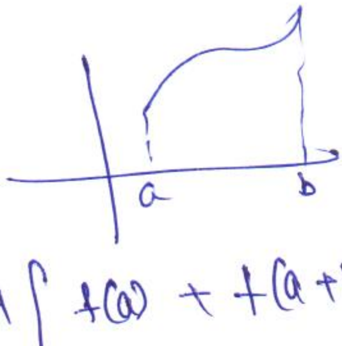
(c) $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

(d) $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$

(e) $\int_0^{2a} f(x)\,dx = \begin{cases} 2\int_0^a f(x)\,dx & \text{if } f(x) = f(2a-x) \\ 0 & \text{if } f(2a-x) = -f(x) \end{cases}$

(f) $\int_{-a}^{a} f(x)\,dx = \begin{cases} 2\int_0^a f(x)\,dx & \text{if } f(-x) = f(x) \text{ even } f^n \\ 0 & \text{if } f(-x) = -f(x) \text{ odd } f^n. \end{cases}$

Limit as a sum



$y = f(x)$ $\quad H = \frac{b-a}{n}$

as $H \to 0,\ n \to \infty$

$$\int_a^b f(x)\,dx = \lim_{H\to 0} H\Big[f(a) + f(a+H) + f(a+2H) + \dots f(a+(n-1)H)\Big]$$

Note:- i) $\lim_{x\to 0} \frac{(e^x - 1)}{x} = 1$

ii) $1+2+3+\dots n = \frac{n(n+1)}{2}$

iii) $1+2+3+\dots(n-1) = \frac{n(n-1)}{2}$

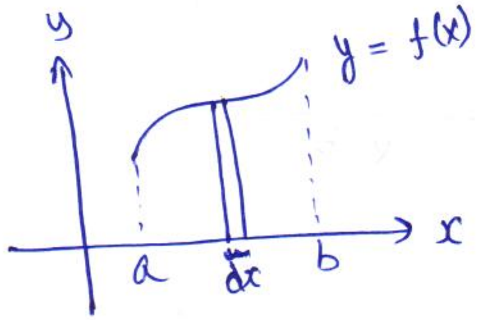
iv) $1^2+2^2+\dots n^2 = \frac{n(n+1)(2n+1)}{6}$

v) $1^2+2^2+\dots(n-1)^2 = \frac{n(n-1)(2n-1)}{6}$

vi) $1^3+2^3+\dots n^3 = \left[\frac{n(n+1)}{2}\right]^2$

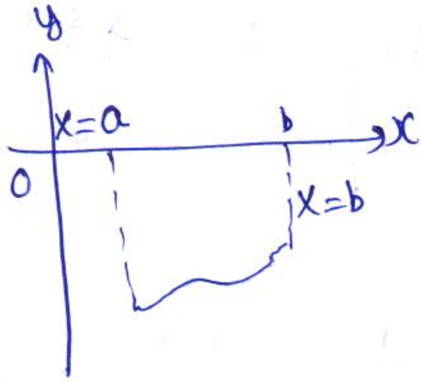
Application of Integrals

(1)



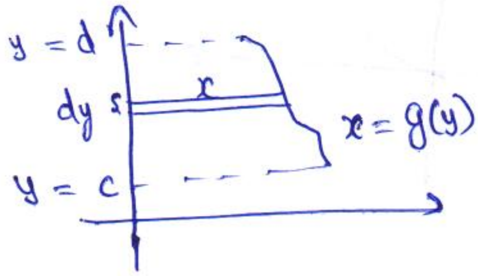
$$\text{Area} = \int_a^b f(x)\,dx$$

(2)



$$\text{Area} = \left| \int_a^b f(x)\,dx \right|$$

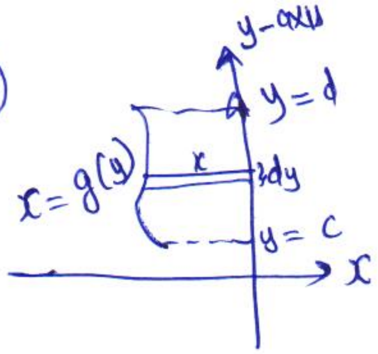
(3)



$$\text{Area} = \int_c^d x\,dy$$

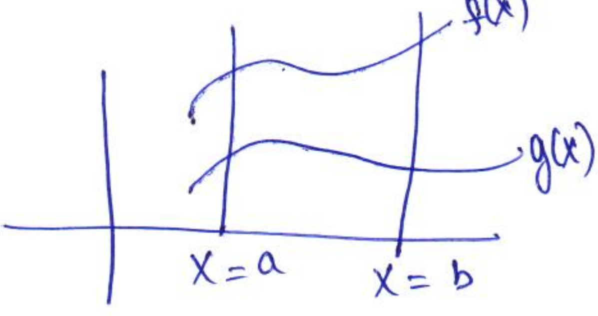
$$\text{Area} = \int_c^d g(y)\,dy$$

(4)



$$\text{Area} = \left| \int_c^d g(y)\,dy \right|$$

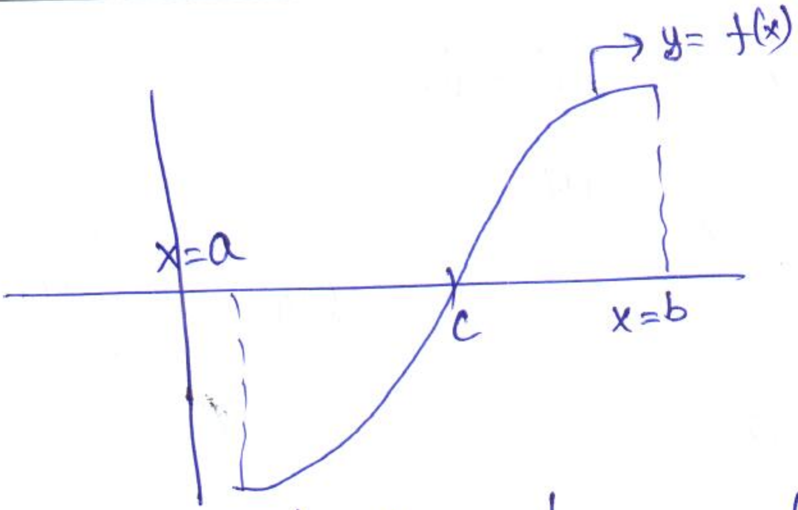
(5)



$$\text{Area} = \int_a^b \left(f(x) - g(x)\right)dx$$

← upper curve – lower curve

6



$$\text{Area} = \left| \int_a^c f(x)\,dx \right| + \int_c^b f(x)\,dx$$

Graphs

$y = x^3$